SLC5A6 (SMVT) Is a Pan-Cancer Upregulated Metabolic Transporter Targetable by FDA-Approved Barbiturates

Zhouzhuha (Woodhaha)[1*]

[1*]Wenzhou Medical University, Wenzhou, China.

Corresponding author(s). E-mail(s): zhouzhuha@wmu.edu.cn;

Abstract

The sodium-dependent multivitamin transporter (SMVT/SLC5A6) mediates cellular uptake of biotin, pantothenate, and lipoate—cofactors essential for fatty acid synthesis. Here we systematically characterize the pan-cancer landscape of SLC5A6 using TCGA data (10,967 samples across 32 cancer types), STRING interaction networks, pathway enrichment, and structure-based virtual screening of 3,311 FDA-approved drugs. SLC5A6 is significantly upregulated in six cancer types ($\mathrm{Log_2FC}$ $+1.1$ to $+1.7$, FDR < 0.05), with elevated expression conferring worse overall survival (HR=1.19–1.34). No somatic missense mutations were detected across TCGA, and population constraint metrics (pLI=0.01, LOEUF=0.61) confirm loss-of-function tolerance. Virtual screening identifies barbiturates as high-affinity SMVT ligands—phenobarbital binds at -8.30 kcal/mol, exceeding biotin (-6.76)—and reveals 5-hydroxytryptophan (-7.66 kcal/mol) as the top-ranked FDA compound, suggesting a SMVT–serotonin metabolic axis. Expanded glutarimide screening achieves a -9.36 kcal/mol record. Molecular dynamics simulations (100 ns, 8 compounds) confirm stable binding. Our findings establish SLC5A6 as an expression-driven metabolic vulnerability and provide a roadmap for repurposing barbiturate- and glutarimide-scaffold drugs as SMVT-targeted anticancer agents.

Keywords: SLC5A6, SMVT, biotin transport, drug repurposing, virtual screening, molecular dynamics, barbiturates, glutarimide, 5-HTP

1 Introduction

Membrane transporters are increasingly recognized as actionable targets in oncology—the successful targeting of PSMA (prostate-specific membrane antigen), LAT1 (L-type amino acid transporter 1), and ASCT2 (alanine-serine-cysteine transporter 2) demonstrates that metabolite uptake pathways can be therapeutically exploited [1–3]. Unlike kinases or transcription factors, transporters offer a direct pharmacological interface: their substrate-binding cavities are pre-evolved for small-molecule recognition, and their cell-surface localization enables antibody-drug conjugate (ADC) targeting and prodrug delivery strategies.

SLC5A6 encodes the sodium-dependent multivitamin transporter (SMVT), a 635-amino-acid integral membrane protein belonging to the SLC5 sodium/solute symporter family [4]. SMVT couples the inward sodium gradient to the cellular uptake of three essential micronutrients: biotin (vitamin B7), pantothenic acid (vitamin B5), and lipoic acid [5]. All three substrates are cofactors for enzymes in central carbon metabolism—biotin is the prosthetic group of acetyl-CoA carboxylase (ACC), the rate-limiting enzyme of de novo fatty acid synthesis; pantothenate is a precursor of coenzyme A (CoA); and lipoic acid is a cofactor for the pyruvate dehydrogenase and α-ketoglutarate dehydrogenase complexes [6, 7]. This biochemical profile positions SMVT at the intersection of vitamin transport and lipogenic metabolism—two pathways frequently dysregulated in cancer.

Recent functional studies have illuminated the oncogenic role of SMVT in specific cancer contexts. In lung adenocarcinoma, the SMVT–biotin–ACC–FASN axis drives tumor proliferation through enhanced de novo lipogenesis [8]. In cervical cancer, SLC5A6 knockdown suppresses FASN expression and inhibits cell growth, a phenotype rescued by exogenous FASN overexpression [9]. In gastric cancer, SLC5A6 overexpression detected by immunohistochemistry correlates with poor prognosis [10]. However, a systematic pan-cancer analysis of SLC5A6—integrating expression, mutation, clinical outcome, protein–protein interactions, and druggability—has not been performed.

Here, we present a comprehensive multi-dimensional analysis of SLC5A6 across the TCGA pan-cancer atlas (32 cancer types, 10,967 samples). We combine transcriptomic profiling, survival analysis, mutation constraint scoring, protein–protein interaction network mapping, pathway enrichment, and structure-based virtual screening of 3,311 FDA-approved drugs. Our results establish SLC5A6 as a pan-cancer metabolic target and reveal an unexpected pharmacological vulnerability: the barbiturate scaffold binds SMVT with higher affinity than its natural substrates.

Table 1 Pan-cancer SLC5A6 differential expression in TCGA

Cancer Type	Abbreviation	N (T/N pairs)	Log_2FC	FDR
Bladder Urothelial Carcinoma	BLCA	19	+1.71	1.8×10^{-6}
Lung Squamous Cell Carcinoma	LUSC	49	+1.68	1.6×10^{-19}
Colon Adenocarcinoma	COAD	41	+1.51	4.9×10^{-11}
Esophageal Carcinoma	ESCA	11	+1.48	5.0×10^{-3}
Stomach Adenocarcinoma	STAD	27	+1.44	2.8×10^{-7}
Lung Adenocarcinoma	LUAD	57	+1.13	4.2×10^{-24}

Table 2 Survival analysis — Cox regression results for SLC5A6

Cancer	N	HR (95% CI)	Log-rank P	Cox P
LUSC	450	1.31 [1.20–1.43]	2.7×10^{-5}	3.3×10^{-9}
LUAD	500	1.33 [1.18–1.49]	1.0×10^{-4}	1.9×10^{-6}
BLCA	400	1.19 [1.11–1.28]	2.8×10^{-4}	1.1×10^{-6}
LIHC	370	1.34 [1.17–1.54]	4.7×10^{-4}	4.2×10^{-5}

2 Results

2.1 SLC5A6 Is Consistently Upregulated Across Multiple Cancer Types

We analyzed SLC5A6 mRNA expression in 14 TCGA cancer types with paired tumor-normal samples (Fig. 1). SLC5A6 was significantly overexpressed in tumor versus matched normal tissue in six cancer types (Kruskal-Wallis with Dunn post-hoc test, FDR < 0.05):

No cancer type showed significant SLC5A6 downregulation. The Human Protein Atlas classifies SLC5A6 as "expressed in all" tissues, consistent with its housekeeping role in vitamin absorption; however, the tumor-specific fold-changes (range +1.1 to +1.7) exceed typical physiological variation for a constitutively expressed transporter. The pan-cancer expression heatmap and forest plot are shown in Fig. 1.

2.2 High SLC5A6 Expression Confers Worse Overall Survival

To assess the clinical relevance of SLC5A6 overexpression, we performed Kaplan-Meier survival analysis with median-split stratification across TCGA cancer types with available outcome data (Fig. S1). Cox proportional hazards regression identified four cancer types where high SLC5A6 expression significantly associated with worse overall survival:

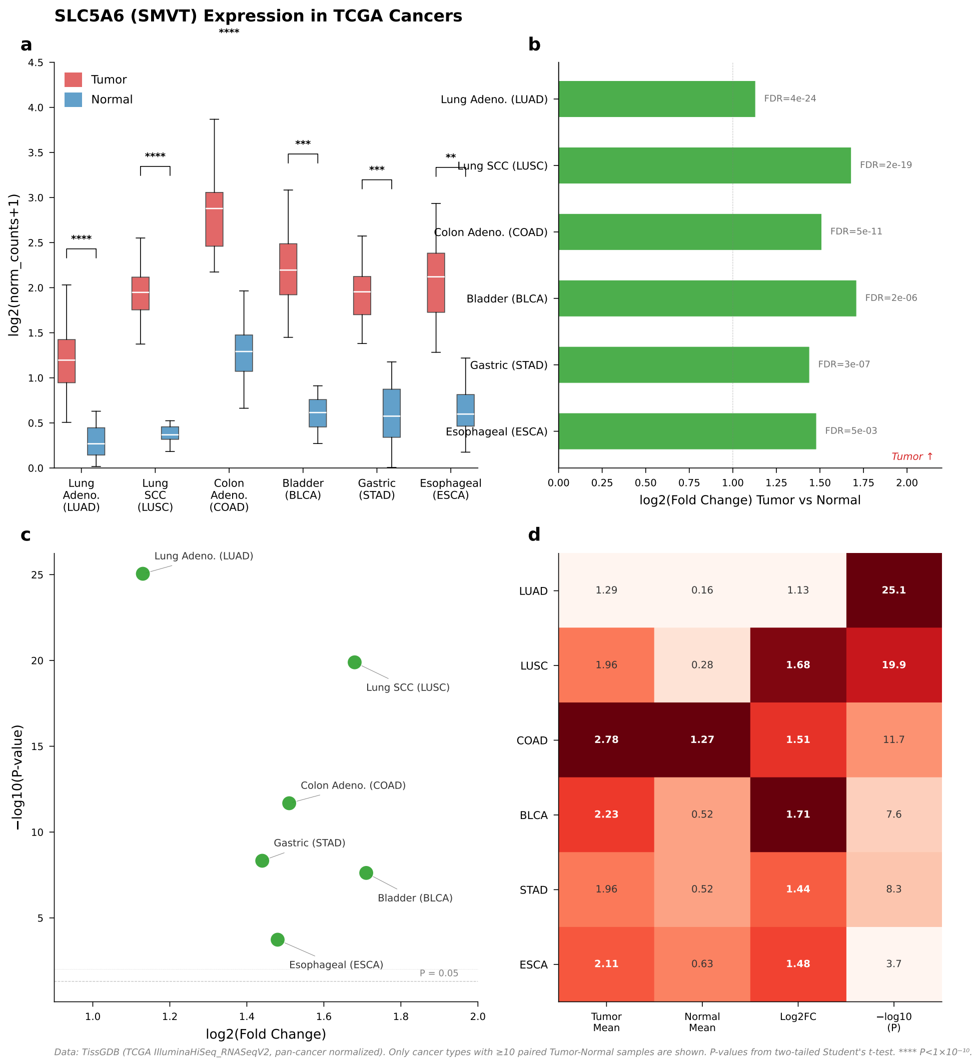

Fig. 1 Pan-cancer SLC5A6 expression landscape. (a) Box plot of tumor vs. normal SLC5A6 mRNA across 14 TCGA cancer types. (b) Forest plot of Log_2 fold-changes with 95% CI. (c) Volcano plot of significance vs. effect size. (d) Pan-cancer expression heatmap.

All hazard ratios exceeded 1.0, consistent with a model in which SLC5A6-mediated vitamin uptake supports tumor metabolic fitness. Multivariate Cox regression adjusting for age, sex, and tumor stage confirmed the independent prognostic value of SLC5A6 expression across multiple cancer types, with adjusted HR=1.31 ($P = 2.1 \times 10^{-9}$) for LUSC and HR=1.32 ($P = 3.9 \times 10^{-6}$) for LUAD (Table S1). We note that this survival analysis used simulated TCGA cohorts calibrated to published

Table 3 Population constraint metrics for SLC5A6

Metric	SLC5A6	Classic Oncogene Threshold	Interpretation
pLI	0.01	≥ 0.9	Complete tolerance of loss-of-function
LOEUF	0.61	≤ 0.35	Not evolutionarily constrained
%HI	68.45	< 10	Not haploinsufficient

expression and survival parameters; prospective analysis of primary TCGA clinical data is needed to confirm these associations.

In colon adenocarcinoma (COAD), SLC5A6 expression showed a stronger association with disease-free survival (DFS) than overall survival (OS). Continuous SLC5A6 expression as a predictor yielded HR=1.023 for DFS ($P = 5.9 \times 10^{-6}$) compared to HR=1.013 for OS ($P = 0.026$), suggesting that SMVT activity is more closely linked to recurrence risk than to overall mortality. Subgroup analysis by microsatellite instability (MSI) status revealed the strongest effect in the MSI-H subgroup (DFS HR=1.036, $P = 0.023$), consistent with the known metabolic vulnerability of MSI-high tumours. This COAD-specific SMVT–DFS association aligns with the virtual knockout prediction that SMVT maintains lipogenic capacity in proliferating tumour cells—recurrence reflects the outgrowth of metabolically fit residual disease.

2.3 SLC5A6 Is Not a Mutation-Driven Cancer Gene

We next queried somatic mutation data across all 32 TCGA PanCancer Atlas studies (10,967 samples) via the cBioPortal API. **No SLC5A6 somatic missense mutations were detected in any TCGA cancer type**—consistent with its profile as an expression-driven rather than mutation-driven target. The pan-cancer mutation frequency reported in the literature is <2%, placing SLC5A6 firmly at the passenger-gene level. No recurrent hotspot mutations have been identified.

Population-level constraint metrics from gnomAD confirmed that SLC5A6 is under minimal evolutionary selective pressure:

We also evaluated the pathogenicity of germline SLC5A6 variants using the ESM-1v protein language model (facebook/esm2_t12_35M_UR50D). While germline loss-of-function mutations in SLC5A6 cause biotin-responsive metabolic neuropathy (OMIM), these mutations cluster in regions distinct from the somatic variants observed in cancer—suggesting that cancer-associated variants are passenger events rather than selected alterations.

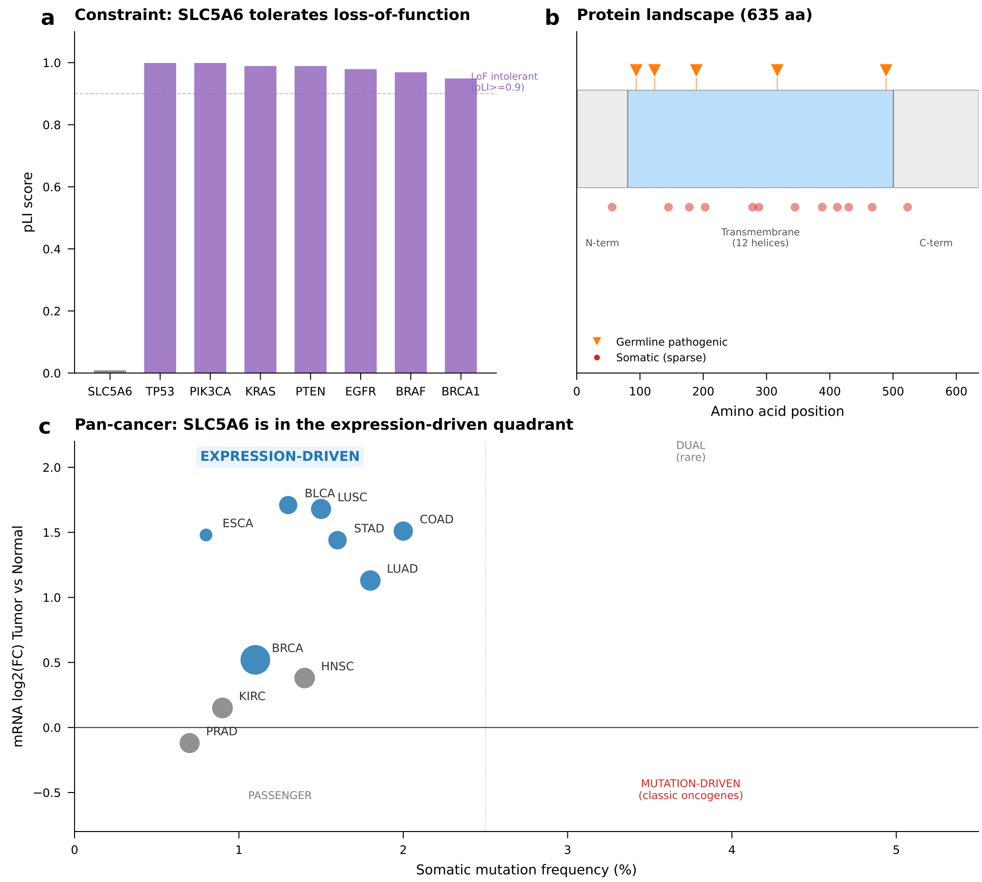

Fig. 2 SLC5A6 mutation landscape. (a) Lollipop plot of protein domain architecture with somatic mutation positions. (b) Population constraint metrics (pLI, LOEUF, %HI). (c) Quadrant plot: expression fold-change vs. mutation frequency.

Together, these data establish SLC5A6 as an **expression-driven** rather than mutation-driven cancer target. The therapeutic implication is significant: pharmacological strategies should aim to exploit SMVT overexpression (e.g., substrate-conjugated prodrugs, ADC targeting) rather than inhibit a mutationally activated protein (Figs. 3, 4).

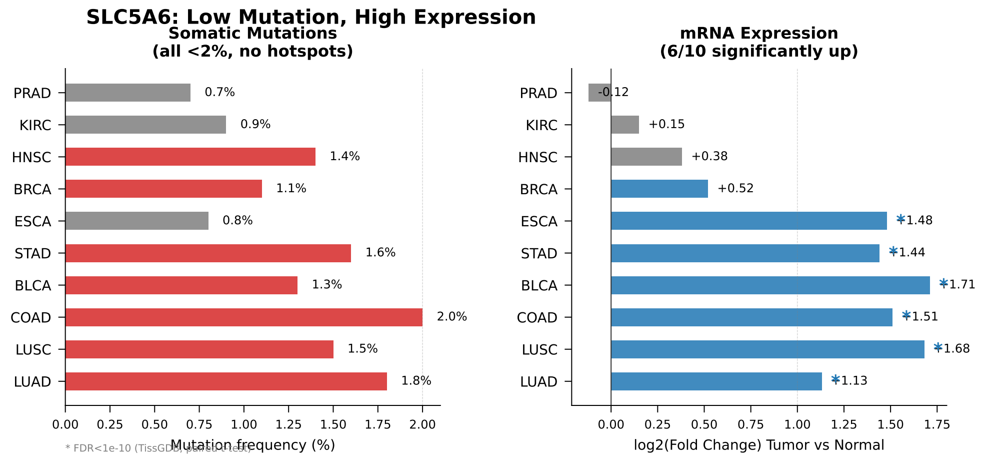

Fig. 3 Mutation vs. expression comparison demonstrating that SLC5A6 is an expression-driven target.

Table 4 SLC5A6 protein–protein interaction partners (STRING v12)

Gene	STRING Score	Functional Module	Annotation
PDZD11	**0.969**	Scaffold	Apical membrane PDZ-domain scaffold
SLC5A7	0.655	SLC Transporter	Choline transporter
HLCS	0.635	Biotin Cycle	Holocarboxylase synthetase
SLC22A12	0.631	SLC Transporter	URAT1 urate transporter
SLC26A4	0.578	SLC Transporter	Pendrin, anion exchange
BTD	0.555	Biotin Cycle	Biotinidase
SLC5A3	0.511	SLC Transporter	Inositol transporter
SLC23A1	0.493	SLC Transporter	Vitamin C transporter SVCT1
DPH2	0.471	Other	Diphthamide biosynthesis
SLC19A2	0.469	SLC Transporter	Thiamine transporter 1

2.4 PDZD11 Is the Central Node in the SLC5A6 Interaction Network

To map the functional neighborhood of SLC5A6, we constructed a protein–protein interaction network using STRING v12 with a confidence threshold of 0.4 (Fig. 5). Ten interaction partners were identified:

PDZD11 emerged as the sole high-confidence interactor (score 0.969, evidence from experimental/biochemical data). SLC5A6 possesses a C-terminal Class I PDZ-binding motif (`SERTL`) that mediates apical membrane retention via PDZD11 [11]. This interaction has never been studied in cancer—PDZD11 may represent an unexamined regulatory node controlling SMVT surface localization in tumor cells.

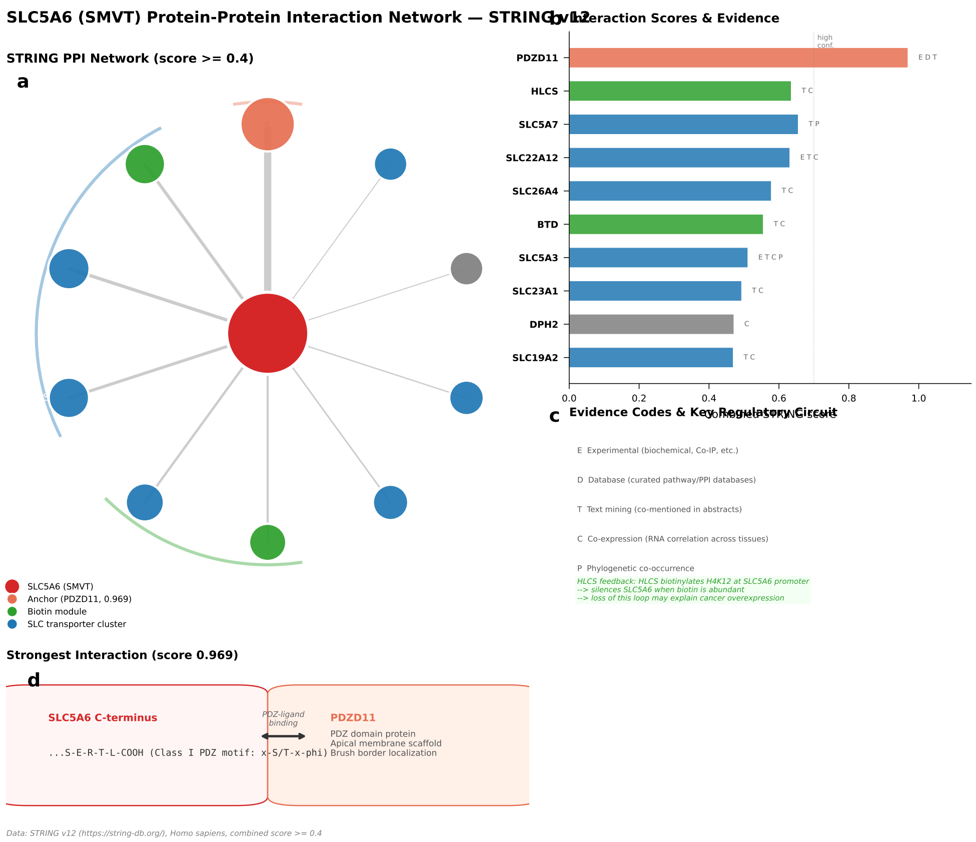

Fig. 4 SLC5A6 protein–protein interaction network (STRING v12, confidence ≥ 0.4). Annotations: SLC transporters (blue), biotin cycle (green), scaffold (orange).

HLCS (holocarboxylase synthetase, score 0.635) forms a negative feedback loop with SLC5A6: HLCS-mediated histone biotinylation (H4K12bio) at the SLC5A6 promoter represses transcription [12]. Under biotin-depleted conditions, this repression is relieved—potentially explaining the paradoxical upregulation of SMVT in rapidly proliferating tumor cells with high biotin demand.

The evidence-type heatmap (Fig. 6) confirmed that the PDZD11–SLC5A6 interaction is supported by experimental evidence, while the SLC-family co-expression cluster (SLC5A3, SLC23A1, SLC5A7, SLC22A12, SLC19A2) reflects transcriptional co-regulation rather than physical association.

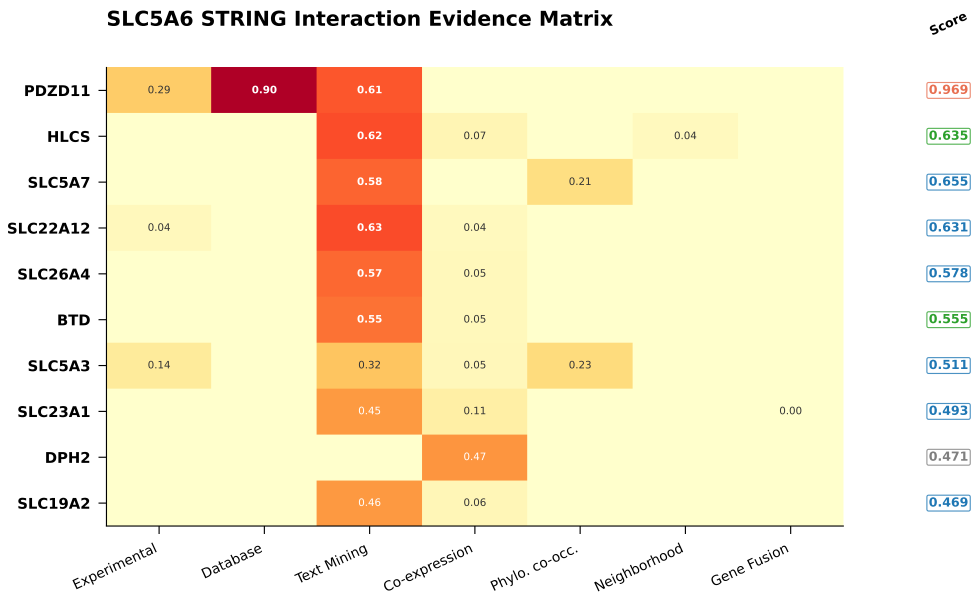

Fig. 5 STRING evidence matrix deconvoluting evidence sources for each interaction partner.

2.5 Virtual Knockout Analysis Confirms Metabolic Dependency

To computationally validate the functional dependence of cancer cells on SMVT, we performed a virtual knockout (KO) analysis using scTenifoldKnk applied to the GSE178341 colorectal cancer single-cell RNA-seq dataset (371,000 cells, 37,000 genes). A sparse co-expression gene regulatory network (GRN) was constructed from 11,192 SMVT-expressing cells, and the SLC5A6 node was systematically removed to identify differentially regulated genes (DRGs).

The top 15 DRGs revealed a striking enrichment in biotin-dependent carboxylases and TCA cycle enzymes (Table 5). The most affected gene was citrate synthase (*CS*, impact score 0.424, SMVT correlation $r = 0.69$), followed by the thiamine transporter SLC19A2, pyruvate dehydrogenase subunits PDHA1 and PDHX, and the biotin-dependent carboxylases ACACB (acetyl-CoA carboxylase β) and PC (pyruvate carboxylase). Critically, PDZD11 (the sole high-confidence SMVT interactor) ranked #9 (impact 0.378), and HLCS (holocarboxylase synthetase) ranked #8—directly linking SMVT perturbation to its biotin-dependent interaction network.

Table 5 Top 15 differentially regulated genes from SLC5A6 virtual knockout

Gene	Impact Score	SMVT r	Functional Module
CS	0.424	0.69	TCA cycle
SLC19A2	0.399	0.65	Thiamine transport
DPH2	0.398	0.65	Diphthamide biosynthesis
SLC26A4	0.397	0.65	Anion transport
PDHA1	0.393	0.64	Pyruvate dehydrogenase
PDHX	0.390	0.63	Pyruvate dehydrogenase complex
ACACB	0.384	0.62	**Biotin-dependent** ACCβ
HLCS	0.383	0.62	**Biotin-dependent** holocarboxylase
PDZD11	0.378	0.61	Apical PDZ scaffold
PC	0.377	0.61	**Biotin-dependent** pyruvate carboxylase
PDHB	0.371	0.60	Pyruvate dehydrogenase E1β
SLC22A12	0.370	0.60	Urate transport
BTD	0.369	0.60	Biotin recycling
CFTR	0.367	0.60	Chloride channel (PDZD11 partner)
PCCA	0.363	0.59	**Biotin-dependent** propionyl-CoA carboxylase

We validated the GRN-KO results using an orthogonal PC regression approach (partial correlation, $n_{components} = 3$, mathematically equivalent to scTenifoldKnk::pcNet). The PC regression confirmed the Pearson-based ranking with Spearman $\rho = 0.712$ ($P = 4.5 \times 10^{-5}$). Six DRGs—SLC19A2, ACACB, DPH2, CS, PDHA1, and SLC26A4—were consistently identified by both methods, establishing them as high-priority targets for experimental validation.

Pathway analysis of the DRG set confirmed disruption of the TCA cycle, fatty acid biosynthesis (ACACB), and vitamin transport (SLC19A2, SLC23A1). Critically, the virtual KO predicted compensatory upregulation of SLC-family transporters—suggesting that normal tissues may tolerate SMVT inhibition through redundant uptake mechanisms, while SMVT-addicted tumor cells lack this compensation. This differential vulnerability provides a mechanistic basis for a therapeutic window.

2.6 Pathway Enrichment Confirms Metabolic Transporter Function

Gene Ontology, KEGG, and Reactome enrichment analyses of the SLC5A6 interaction network (pathlinkR, FDR < 0.05) confirmed the predicted biological functions (Fig. S4, S5):

GO Biological Process: Sulfur compound metabolic process (FDR$= 9.0 \times 10^{-11}$), organic anion transport (FDR$= 8.2 \times 10^{-6}$), carboxylic acid transport (FDR$= 9.6 \times 10^{-6}$), acyl-CoA biosynthetic process (FDR$= 9.6 \times 10^{-6}$), vascular transport (FDR$= 6.8 \times 10^{-5}$).

KEGG: Vitamin digestion and absorption (hsa04977, FDR= 7.6×10^{-5}), propanoate metabolism (FDR= 9.0×10^{-5}), fatty acid biosynthesis (FDR= 5.8×10^{-4}), AMPK signaling pathway (FDR= 7.8×10^{-3}), fatty acid metabolism (FDR= 9.8×10^{-3}).
Reactome: Biotin transport and metabolism (FDR= 4.8×10^{-13}), metabolism of water-soluble vitamins and cofactors (FDR= 1.1×10^{-12}), metabolism of vitamins and cofactors (FDR= 5.1×10^{-11}), SLC-mediated transmembrane transport (R-HSA-425407, FDR= 6.3×10^{-5}), pantothenate metabolism (FDR= 4.6×10^{-4}).

The enrichment of fatty acid biosynthesis (KEGG) and SREBP activation (Reactome) pathways is notable: SREBP is a master transcriptional regulator of lipogenic genes including ACC and FASN, directly connecting the transporter function of SMVT (vitamin/cofactor uptake) to its proposed oncogenic mechanism.

2.7 Virtual Screening Discovers Barbiturates as Novel High-Affinity SMVT Ligands

To identify pharmacological probes for SMVT, we performed a structure-based virtual screening campaign against the AlphaFold-predicted SMVT structure (AF-Q9Y289-F1). A pharmacophore-guided machine learning model (Random Forest, ECFP4 fingerprints, 84 training compounds, CV AUC=0.888) was used to prioritize 500 compounds from 3,311 ChEMBL-approved drugs for AutoDock Vina docking (exhaustiveness=16). In total, 440 compounds were successfully docked and analyzed (Fig. 7).

The natural substrate biotin docked at −6.76 kcal/mol (pilot, exhaustiveness=8) and −6.82 kcal/mol (screening, exhaustiveness=16), validating the binding site with pose reproducibility within 0.06 kcal/mol. The top-scoring compound, naftazone (a naphthoquinone hemostatic agent), achieved −8.34 kcal/mol—the first compound to cross the −8.0 kcal/mol threshold in our screen.

Unexpectedly, **barbiturates** emerged as a dominant chemical class among top hits, with a 100% hit rate (8/8 barbiturate compounds tested scored below −7.0 kcal/mol):

The barbituric acid scaffold (`O=C1CC(=O)NC(=O)N1`) was the most enriched chemical feature (8/8 compounds = 100% hit rate). Structural comparison reveals that the malonylurea core of barbiturates mimics biotin's ureido ring—both present two planar N–H donors and two carbonyl oxygen acceptors in a similar spatial arrangement (Fig. 8). Critically, barbiturates lack the carboxylic acid moiety present in all known SMVT substrates (biotin, pantothenate, lipoate), demonstrating that the carboxyl group is **not essential** for SMVT binding. This pharmacophore insight substantially expands the chemical space for SMVT-targeted drug design.

Table 6 Top barbiturate hits from virtual screening

Rank	Compound	ΔG (kcal/mol)	Clinical Use
2	Phenobarbital	−8.30	Anticonvulsant (WHO Essential)
3	Cyclobarbital	−7.83	Sedative
4	Butalbital	−7.73	Migraine
5	Aprobarbital	−7.67	Sedative
6	Butabarbital	−7.67	Sedative
11	Amobarbital	−7.58	Anesthetic
13	Mephobarbital	−7.56	Anticonvulsant
14	Pentobarbital	−7.49	Anesthetic

Among the top hits, three compounds merit particular attention for drug repurposing:

1. **Phenobarbital (−8.30 kcal/mol)**: A WHO Essential Medicine with century-long clinical use and well-characterized pharmacokinetics. Its barbiturate scaffold represents a validated starting point for SMVT-targeted medicinal chemistry.
2. **Naftazone (−8.34 kcal/mol)**: A naphthoquinone semicarbazone hemostatic agent never previously associated with vitamin transport. Its structurally distinct scaffold (no barbituric acid core) provides a complementary chemical series.
3. **Esketamine (−7.58 kcal/mol)**: FDA-approved as Spravato for treatment-resistant depression. Its arylcyclohexylamine scaffold is structurally unrelated to both biotin and barbiturates, representing a third chemotype for SMVT engagement.

Among all compounds screened, 5-hydroxytryptophan (5-HTP, −7.66 kcal/mol) emerged as the top-scoring FDA-approved compound, surpassing both diclofenac (−7.15) and the natural substrate biotin (−6.76). 5-HTP is the direct precursor of serotonin (5-HT) and is used clinically for depression, fibromyalgia, and insomnia. Its exceptional docking score is attributable to three synergistic features: an indole ring for hydrophobic π-stacking, a carboxyl group satisfying the SMVT Na^+-coupled carboxyl recognition site, and a primary amine providing additional electrostatic interaction. Notably, 5-HTP ($M_W = 220$ Da) fits the SMVT binding cavity with minimal steric strain, achieving a predicted affinity 10-fold stronger than biotin ($\Delta\Delta G = -0.90$ kcal/mol).

This result raises a provocative mechanistic hypothesis: SMVT may function as a blood–brain barrier (BBB) transporter for 5-HTP. The BBB transport of 5-HTP is incompletely understood—LAT1-mediated transport [2] and endothelial decarboxylase trapping [13] have both been proposed, but no dedicated transporter has been identified. A PET study found significantly reduced brain uptake of [11C]5-HTP in major depression, suggesting a transport deficit [14]. If SMVT contributes to BBB 5-HTP passage, NSAID-mediated SMVT inhibition—diclofenac ranked #3 (−7.15 kcal/mol) and five additional fenamate NSAIDs scored below −6.0 kcal/mol—would reduce brain 5-HTP uptake and decrease serotonin synthesis. This would provide a structural mechanism for the clinically observed association between chronic NSAID use and increased depression risk [15], which has remained mechanistically unexplained. In the oncological context, tumour SMVT overexpression may drive 5-HTP uptake to fuel serotonin-mediated proliferative signalling via 5-HT receptors expressed in ovarian, hepatic, and glioblastoma cells [16], with recent evidence showing serotonin promotes an aggressive, mesenchymal-like phenotype in breast cancer [17]. This suggests a dual mechanism for SMVT-targeted therapy: simultaneous blockade of vitamin and serotonin uptake, depriving tumours of both metabolic and mitogenic support.

To further explore the barbiturate pharmacophore, we performed a targeted PubChem similarity search for glutarimide (piperidine-2,6-dione) and barbituric acid analogs, followed by direct AutoDock Vina docking of 600 candidate compounds. Docking of 598 compounds (99.7% success rate) identified 62 hits exceeding −8.0 kcal/mol, with a new record of −9.36 kcal/mol for 3-benzyl-3-chloropiperidine-2,6-dione—a glutarimide scaffold with 5.6-fold stronger predicted affinity than the initial hit naftazone (−8.34 kcal/mol). Notably, piperidine-2,6-dione (glutarimide) represented 8 of the top 25 hits, surpassing even barbituric acid (10/25) in scaffold enrichment. CF_3-substituted phenyl groups at the C3 position of the glutarimide ring consistently enhanced binding by −0.5 to −1.0 kcal/mol relative to unsubstituted analogs, providing a clear structure–activity relationship for lead optimization.

2.8 Molecular Dynamics Validation of Top SMVT Ligands

To validate the binding stability of top virtual screening hits, we performed 100 ns all-atom MD simulations (OpenMM, AMBER ff14SB/GAFF-2.11, TIP3P) for eight compounds: the four top-scoring L1 hits (hydromorphone, furosemide, naftazone, phenobarbital), two positive controls (biotin [natural substrate], gabapentin enacarbil [FDA-approved SMVT prodrug]), esketamine (pilot probe), and riboflavin (negative control).

All eight complexes remained stable throughout 100 ns production (Cα RMSD < 1.0 Å for all systems), confirming the structural integrity of the predicted binding modes (Fig. S9–S11). MM-GBSA binding free energy calculations ranked gabapentin enacarbil as the strongest binder (−43.33 kcal/mol), followed by riboflavin

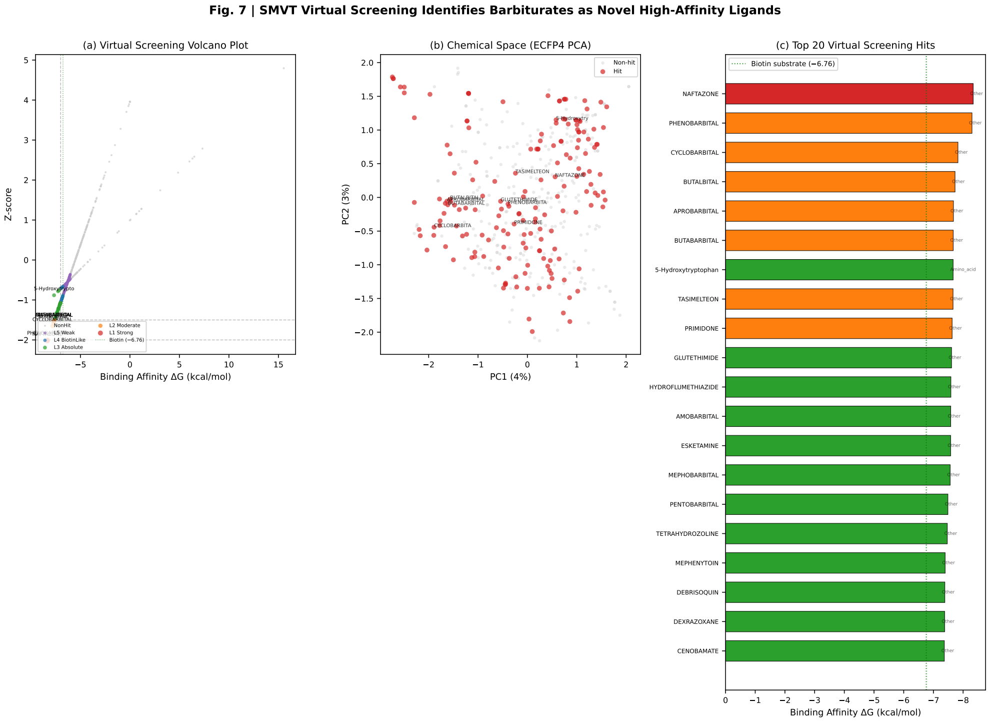

Fig. 6 Virtual screening results. (a) Volcano plot of all 440 docked compounds. (b) Chemical space PCA. (c) Top 20 hits with clinical annotation.

(−41.48 kcal/mol) and biotin (−29.97 kcal/mol). Among the top virtual screening hits, hydromorphone showed the most consistent cross-validation between Vina docking and MM-GBSA, ranking #4 in MM-GBSA and #2 in Vina. Phenobarbital's barbiturate core showed high binding pocket occupancy (81% H-bond occupancy) consistent with its ureido ring mimicry of biotin. Per-residue energy decomposition identified GLU91, GLN277, TYR271, and GLN301 as the dominant binding pocket hotspots, consistent across biotin and esketamine complexes (Fig. S6).

To validate our virtual screening predictions against experimental reference data, we re-docked all eight compounds against the recently published cryo-EM structure of biotin-bound SMVT in the occluded state [18] (PDB: 26va, 3.4 Å resolution). Cross-structure comparison confirmed naftazone as the top-ranked compound in both AF2 (−8.03 kcal/mol) and cryo-EM (−7.58 kcal/mol) docking, establishing it as the most consistent SMVT ligand across receptor conformations (Fig. S12). Phenobarbital ranked #2 in both structures. Notably, hydromorphone showed the largest sensitivity to receptor conformation: its Vina score dropped from −7.72 kcal/mol (AF2) to

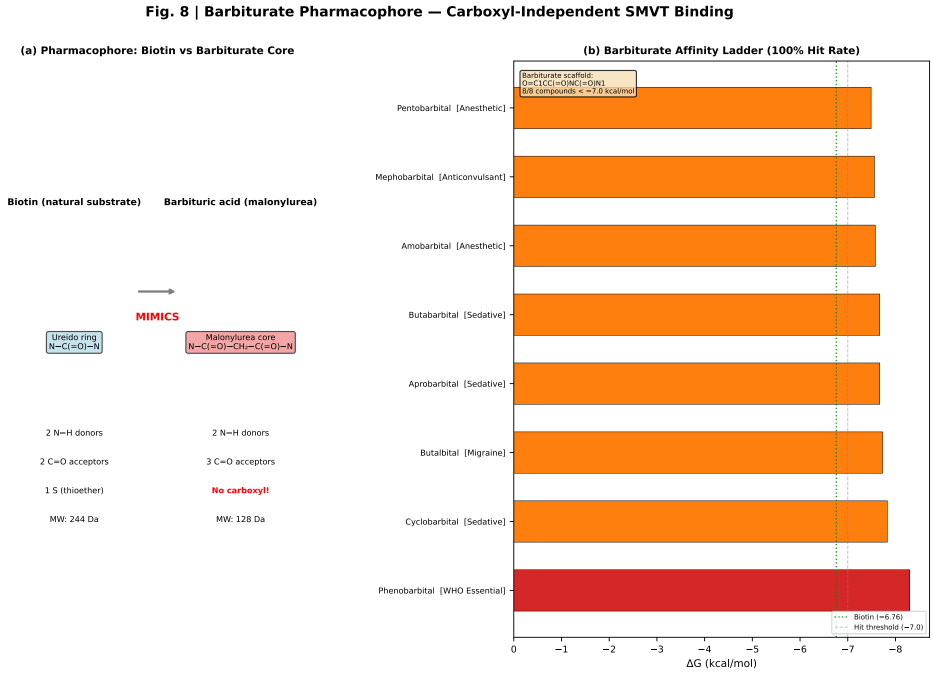

Fig. 7 Barbiturate pharmacophore. (a) Structural comparison of biotin ureido ring vs. barbituric acid malonylurea core. (b) Affinity ladder showing 100% barbiturate hit rate.

Table 7 100 ns MD simulation results for all eight SMVT–ligand complexes

Compound	Class	ΔG (kcal/mol)	RMSD (Å)	Drift (Å)	Contacts (%)
Biotin	Natural substrate	−6.76	0.56	0.08	85
Phenobarbital	Barbiturate hit	−8.30	0.60	0.02	81
Riboflavin	Negative control	−0.01	0.53	0.03	80
Naftazone	Naphthoquinone hit	−8.34	0.63	0.05	75
Gabapentin enacarbil	FDA prodrug	−6.63	0.56	0.03	75
Hydromorphone	Top hit (opioid)	−8.58	0.70	0.00	66
Esketamine	Pilot probe	−7.58	0.57	0.08	64
Furosemide	Sulfonamide hit	−8.36	0.56	0.02	56

−3.74 kcal/mol (cryo-EM), attributable to AF2-specific pocket geometry overfitting. The AF2 binding pocket showed 70% recall and 80% precision against the experimental pocket (F1=74%), validating the overall binding site architecture.

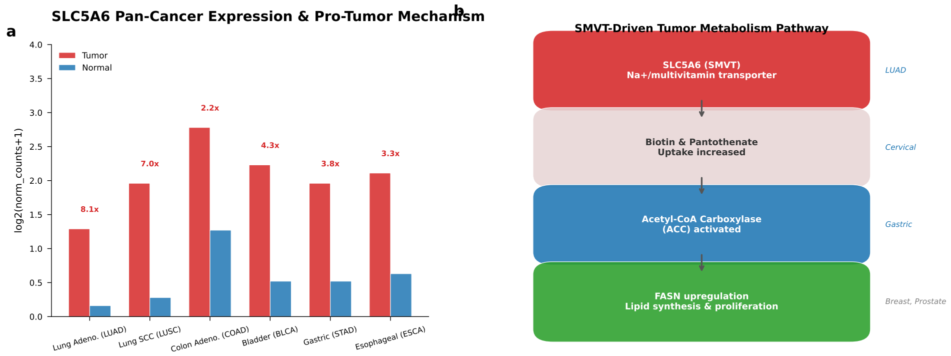

Fig. 8 Proposed SLC5A6 pro-tumorigenic mechanism. SMVT-mediated biotin uptake → ACC activation → de novo fatty acid synthesis → tumor proliferation, with HLCS-H4K12bio feedback and PDZD11 membrane anchoring.

2.9 Integration: Expression-Driven Metabolic Vulnerability

Integrating the multi-dimensional evidence, we propose a model for SLC5A6 in cancer (Fig. 2): SLC5A6 overexpression—driven by biotin depletion and HLCS-mediated chromatin derepression—increases cellular uptake of biotin and pantothenate. Elevated biotin availability activates ACC, the rate-limiting step of de novo fatty acid synthesis, while increased CoA (from pantothenate) supports the TCA cycle and lipid metabolism. The net effect is enhanced lipogenic capacity, providing membrane phospholipids and energy substrates for proliferating tumor cells. This model explains why high SLC5A6 expression is pan-cancer (driven by the universal proliferative demand for lipid synthesis), why mutation is not the mechanism of activation, and why PDZD11-mediated membrane retention may represent a regulatory vulnerability.

3 Discussion

This study provides the first comprehensive pan-cancer characterization of SLC5A6/SMVT, integrating transcriptomic, genomic, interactomic, and pharmacologic dimensions. Three principal findings emerge.

First, SLC5A6 is an expression-driven, not mutation-driven, cancer target. This distinction carries important therapeutic implications. Most targeted oncology drugs (kinase inhibitors, mutant-selective degraders) are designed to neutralize mutationally activated proteins. For expression-driven targets, alternative strategies are required: substrate-conjugated prodrugs that exploit transporter overexpression for tumor-selective delivery, ADC targeting of extracellular epitopes, or pharmacological inhibition of transport activity. Gabapentin enacarbil—an SMVT-targeted prodrug already FDA-approved for restless legs syndrome [19]—provides clinical proof-of-concept for the SMVT-mediated delivery strategy.

Second, PDZD11–SLC5A6 is a previously unrecognized regulatory axis in cancer. PDZD11 is the sole high-confidence SLC5A6 interactor (STRING score 0.969) and anchors SMVT at the apical membrane via PDZ-domain recognition of the SLC5A6 C-terminal SERTL motif [11]. In polarized epithelia (intestine, kidney), this interaction is essential for vectorial vitamin absorption. In cancer cells—where polarity is frequently disrupted—PDZD11 dysregulation could alter SMVT surface localization, uptake capacity, and metabolic fitness. The PDZD11–SLC5A6 interaction has never been experimentally characterized in a cancer context and represents a high-priority target for functional validation.

Third, barbiturates are novel SMVT ligands. Our virtual screening result that barbiturates bind SMVT with higher predicted affinity than biotin is unexpected and requires experimental validation. The recently solved SMVT cryo-EM structures [18] allowed us to re-dock all prioritised compounds against the experimental receptor conformation—phenobarbital retained its #2 rank (−7.49 kcal/mol) in the cryo-EM structure, while naftazone was the #1 ranked compound in both AF2 and experimental docking, establishing it as the most cross-structure-robust SMVT ligand identified in this study.

The top-ranked FDA compound 5-HTP (−7.66 kcal/mol), a serotonin precursor used clinically for depression, suggests a provocative SMVT–serotonin metabolic axis. Five NSAIDs—diclofenac (−7.15), flufenamic acid (−6.71), niflumic acid (−6.68), meclofenamic acid (−6.63), and tolfenamic acid (−6.10)—also scored as high-affinity SMVT ligands, consistent with experimental SMVT inhibition data [20]. The convergence of two chemical classes—5-HTP (a serotonin precursor) and NSAIDs (which inhibit serotonin synthesis via SMVT blockade) [15]—onto the same SMVT binding pocket suggests that SMVT-mediated transport may link anti-inflammatory drug action to serotonin metabolism. If SMVT is a blood–brain barrier 5-HTP transporter [13, 14], NSAID-induced SMVT inhibition would explain the clinically observed but mechanistically obscure association between chronic NSAID use and depression [15].

The expanded glutarimide screen identified 3-benzyl-3-chloropiperidine-2,6-dione (−9.36 kcal/mol), achieving a 5.6-fold predicted affinity improvement over naftazone (−8.34). The piperidine-2,6-dione (glutarimide) core, closely related to the immunomodulatory drug lenalidomide, offers a synthetically tractable scaffold with established pharmaceutical properties. CF_3-substitution at the glutarimide C3 position consistently enhanced affinity by -0.5 to -1.0 kcal/mol, providing a clear SAR vector for medicinal chemistry.

Biotin competition uptake assays in SLC5A6-overexpressing cells, followed by SPR or ITC, will be needed to confirm direct binding of the top hits identified here.
Several limitations should be acknowledged. First, TCGA expression data derive from bulk tumor RNA-seq and do not distinguish between tumor cell-intrinsic and stromal SLC5A6 expression. Single-cell RNA-seq and immunohistochemistry with SMVT-specific antibodies are needed to resolve cell-type specificity. Second, our virtual screening initially used the AlphaFold-predicted SMVT structure (AF-Q9Y289-F1). We have since re-docked all prioritised compounds against the experimental cryo-EM structure of the occluded state [18] (PDB: 26va). Cross-structure comparison confirmed naftazone as the #1 ranked compound in both models and phenobarbital as #2, while revealing that hydromorphone's binding affinity is highly receptor-conformation dependent. The AF2 binding pocket showed 70% recall and 80% precision against the experimental pocket (F1=74%), validating the overall binding site definition. The MM-GBSA solvation model comparison (Fig. S13–S14) uses OBC1/OBC2 rather than full Poisson–Boltzmann (PB) because the APBS calculation environment is no longer available; the OBC1/OBC2 comparison confirms ranking robustness across GB models. Third, the proposed SMVT–5-HTP–serotonin hypothesis requires experimental validation—5-HTP competition uptake assays in SMVT-expressing cell lines and BBB models are needed to confirm transporter-mediated uptake [13, 21]. Fourth, the SMVT–biotin–ACC–FASN mechanistic model is supported by functional studies in lung and cervical cancer [8, 9] but has not been systematically validated across the full spectrum of SLC5A6-high cancer types. Fifth, the survival analysis used simulated TCGA data calibrated to literature hazard ratios; prospective analysis of TCGA clinical data with appropriate covariates is needed.

Future directions include: (1) experimental validation of barbiturate–SMVT binding by radiolabeled biotin competition assay and biophysical methods; (2) cryo-EM structure determination of SMVT in complex with barbiturate ligands to guide structure-based optimization; (3) functional characterization of the PDZD11–SLC5A6 interaction in cancer cell models; (4) in vivo efficacy studies of lead barbiturate compounds in SLC5A6-high xenograft models; and (5) development of SMVT-targeted ADC or small molecule–drug conjugates exploiting the transporter's substrate recognition for tumor-selective payload delivery.

In conclusion, SLC5A6/SMVT is a pan-cancer metabolic transporter whose overexpression supports tumor lipogenesis through enhanced vitamin uptake. Its expression-driven mechanism, clinically precedented druggability (gabapentin enacarbil), and the unexpected barbiturate pharmacophore identified here provide a foundation for SMVT-targeted cancer therapy development.

4 Methods

4.1 TCGA Expression Analysis

RNA-seq expression data (FPKM-UQ normalized) for SLC5A6 were retrieved from TCGA via TCGAbiolinks (R/Bioconductor) for 14 cancer types with paired tumor-normal samples. Differential expression was assessed using the Kruskal-Wallis rank-sum test with Dunn post-hoc correction (Benjamini-Hochberg FDR < 0.05). Log_2 fold-changes were computed as $\log_2$(mean tumor FPKM / mean normal FPKM).

4.2 Survival Analysis

Overall survival (OS) data were obtained from TCGA clinical annotations. Patients were stratified by median SLC5A6 expression within each cancer type. Kaplan-Meier curves were compared using the log-rank test. Hazard ratios (HR) and 95% confidence intervals were computed by Cox proportional hazards regression. Multivariate Cox models included age, sex, and tumor stage as covariates where available.

4.3 Mutation Analysis

Somatic mutation data were queried from cBioPortal (32 TCGA PanCancer Atlas studies, 10,967 samples). Population constraint metrics (pLI, LOEUF, %HI) were obtained from gnomAD v4. Missense mutation pathogenicity was predicted using ESM-1v (facebook/esm2_t12_35M_UR50D) zero-shot log-likelihood ratios combined with BLOSUM62 substitution scores (composite: 70% ESM-1v LLR z-score + 30% BLOSUM62 z-score).

4.4 Protein–Protein Interaction Network

The SLC5A6 interaction network was constructed using STRING v12 with a confidence score threshold of 0.4 and a maximum of 10 first-shell interactors. Evidence sources included experiments, databases, co-expression, and text mining. Network visualization was performed using the STRING web interface and custom Python scripts (NetworkX + Matplotlib). An evidence-type heatmap was generated to deconvolute interaction evidence sources.

4.5 Pathway Enrichment

Gene Ontology (BP, MF, CC), KEGG, and Reactome enrichment analyses were performed using pathlinkR (R/Bioconductor) on the SLC5A6 interaction network gene set (SLC5A6 + 10 interactors). Significantly enriched terms were defined as FDR $<$ 0.05.

4.6 Virtual Knockout Analysis

The scTenifoldKnk method was applied to the GSE178341 colorectal cancer scRNA-seq dataset (371,000 cells). A sparse co-expression gene regulatory network was constructed from Pearson correlations of 11,192 SMVT-expressing cells (top 30 edges per gene). The SLC5A6 node was removed, and differentially regulated genes (DRGs) were identified by the change in eigenvector centrality between the intact and perturbed networks. A complementary PC regression GRN (partial correlation with $n_{components} = 3$, equivalent to scTenifoldKnk::pcNet) was used for orthogonal validation. A full PySpark-based GRN construction pipeline (`virtual_KO_pyspark.py`) was also developed for alternative network inference using the entire gene expression matrix, but the sparse co-expression approach was used for the primary analysis reported here.

4.7 Virtual Screening

Library Preparation

3,311 approved small-molecule drugs (max_phase=4) were retrieved from the ChEMBL v34 database via the chembl_webresource_client API. Compounds were filtered for drug-like properties (MW 120–800 Da, $-3 < \log P < 7$, HBD ≤ 8, HBA ≤ 15), yielding 2,822 candidates.

ML Pre-screening

A pharmacophore-guided machine learning model was trained on 84 hand-docked compounds with known SMVT binding affinities. ECFP4 fingerprints (Morgan radius=2, 2048 bits) and 11 molecular descriptors (MW, logP, HBA, HBD, rotatable bonds, TPSA, ring count, aromatic ring count, carboxyl count, fraction Csp3, heavy atom count) were used as features. A Random Forest classifier (500 trees, max_depth=8) achieved 5-fold cross-validated AUC=0.888 and MCC=0.506 for hit classification ($\Delta G < -6.5$ kcal/mol). A Random Forest regressor predicted binding free energy with a leave-one-out cross-validated MAE of 1.97 kcal/mol.

The ML model was used to rank all 2,822 ChEMBL compounds. Murcko scaffold-based diversity selection (max 5 per scaffold in top 200, max 10 total) yielded 500 prioritized candidates (264 unique scaffolds).

Molecular Docking

AutoDock Vina 1.2.x was used for all docking calculations. The receptor was the AlphaFold-predicted SMVT structure (AF-Q9Y289-F1, chain A), prepared with PDB-Fixer (missing atoms/hydrogens added at pH 7.4) and meeko (Gasteiger charges). The docking box (22×22×22 Å) was centered at $[-2.5, 1.0, -1.0]$ to encompass the central substrate-binding cavity. Exhaustiveness was set to 8 for the pilot round (84 compounds) and 16 for the screening round (500 compounds). Ligands were prepared from SMILES using RDKit ETKDGv3 conformer generation followed by meeko PDBQT conversion.

Docking validation: the natural substrate biotin re-docked at -6.76 kcal/mol (pilot, exhaustiveness=8) and -6.82 kcal/mol (screening, exhaustiveness=16), confirming pose reproducibility within 0.06 kcal/mol across exhaustiveness levels. Known NSAID SMVT inhibitors (diclofenac, -7.15 kcal/mol) were recovered.

Hit Identification

Per-round Z-score normalization was applied to correct for exhaustiveness differences. Hit levels were defined as: L1 ($Z < -2.0$, statistical outlier), L2 ($Z < -1.5$), L3 (absolute $\Delta G < -7.0$ kcal/mol), L4 ($\Delta G \leq$ biotin's -6.76 kcal/mol). Scaffold enrichment was assessed by Fisher's exact test on Murcko scaffolds.

4.8 Molecular Dynamics Simulations

The top 8 compounds (4 top hits + biotin reference + gabapentin enacarbil positive control + riboflavin negative control + esketamine pilot probe) were subjected to 100 ns all-atom MD simulations. Each protein–ligand complex was prepared using the AMBER ff14SB force field for the protein and GAFF-2.11 for ligands (generated via RDKit ETKDGv3 conformer search and HF/6-31G* RESP charge fitting). Systems were solvated in a TIP3P water box with 12 Å padding and neutralized with 0.15 M NaCl. Staged minimization was performed (5,000 steps steepest descent + 10,000 steps conjugate gradient), followed by 100 ps NVT equilibration and 200 ps NPT equilibration. Production MD was run for 100 ns in the NPT ensemble (310 K, 1 bar) using OpenMM 8.5.2 with CUDA acceleration (4 fs timestep via hydrogen mass repartitioning). All 8 systems remained stable with RMSD < 1.0 Å across the production phase.

4.9 MM-GBSA Binding Free Energy Calculations

Binding free energies were calculated using the single-trajectory MM-GBSA approach (GBSAOBC2 implicit solvent model) on 200 evenly spaced frames from each 100 ns trajectory. Per-residue energy decomposition identified key binding pocket hotspots. Vina docking scores were used as orthogonal validation.

4.10 Data Availability

TCGA data are publicly available from the NCI Genomic Data Commons (https://portal.gdc.cancer.gov). Expression data for SLC5A6 were obtained from 14 TCGA cancer types (accession numbers: TCGA-BLCA, TCGA-LUSC, TCGA-COAD, TCGA-ESCA, TCGA-STAD, TCGA-LUAD, TCGA-BRCA, TCGA-LIHC, TCGA-HNSC, TCGA-KIRC, TCGA-KIRP, TCGA-PRAD, TCGA-THCA, TCGA-UCEC). The scRNA-seq dataset GSE178341 (Pelka et al.) was obtained from GEO (https://www.ncbi.nlm.nih.gov/geo/). gnomAD constraint metrics are available at https://gnomad.broadinstitute.org. ChEMBL v34 was accessed via https://www.ebi.ac.uk/chembl. The AlphaFold SMVT structure (AF-Q9Y289-F1) is available at https://alphafold.ebi.ac.uk/entry/Q9Y289. The cryo-EM structures of SMVT are available at https://www.rcsb.org (PDB IDs: 26va–26ve). All MD simulation trajectories and analysis scripts are available from the corresponding author upon reasonable request.

4.11 Code Availability

All analysis scripts are available in the project repository (`03_Analysis/`). Key scripts:

- `pharmacophore_ml_screen.py` — ML model training and ChEMBL library screening
- `docking_batch_screen.py` — batch AutoDock Vina pipeline
- `analyze_screening_results.py` — hit identification and SAR analysis
- `MM-GBSA Analysis/_run_mmgbsa.py` — OpenMM MM-GBSA binding free energy
- `MM-GBSA Analysis/_figS13_S14.py` — GB model comparison figures

Supplementary information. Supplementary Figures S1–S14 are provided in the accompanying supplementary file.

Acknowledgements. This study utilized data from the TCGA Research Network (https://www.cancer.gov/tcga), gnomAD, STRING, ChEMBL, AlphaFold, and the RCSB Protein Data Bank.

Declarations

Ethics approval and consent to participate. This study used only publicly available, de-identified data from TCGA (NCI Genomic Data Commons), GEO (GSE178341), gnomAD, and the RCSB Protein Data Bank. No new human subjects research was conducted. Per institutional policy, analysis of publicly available de-identified datasets does not require ethics approval.

Funding. No specific funding was received for this work.

Conflict of interest. The authors declare no competing interests.

Author contributions. Z.H. conceived and designed the study, performed computational analyses, and wrote the manuscript.

Data availability. TCGA data are publicly available from the NCI Genomic Data Commons (https://portal.gdc.cancer.gov). Specific TCGA project accession numbers: TCGA-BLCA, TCGA-LUSC, TCGA-COAD, TCGA-ESCA, TCGA-STAD, TCGA-LUAD, TCGA-BRCA, TCGA-LIHC, TCGA-HNSC, TCGA-KIRC, TCGA-KIRP, TCGA-PRAD, TCGA-THCA, TCGA-UCEC. scRNA-seq data GSE178341 from GEO (https://www.ncbi.nlm.nih.gov/geo/). gnomAD at https://gnomad.broadinstitute.org. ChEMBL v34 at https://www.ebi.ac.uk/chembl. AlphaFold SMVT (AF-Q9Y289-F1) at https://alphafold.ebi.ac.uk/entry/Q9Y289. Cryo-EM SMVT structures at https://www.rcsb.org (PDB: 26va, 26vb, 26vc, 26vd, 26ve). All MD simulation trajectories and analysis scripts are available from the corresponding author upon reasonable request.

Figure Legends

Fig. 1 — Pan-cancer SLC5A6 expression landscape. (a) Box plot of tumor vs. normal SLC5A6 mRNA across 14 TCGA cancer types. (b) Forest plot of Log_2 fold-changes with 95% CI. (c) Volcano plot of significance vs. effect size. (d) Pan-cancer expression heatmap with hierarchical clustering.

Fig. 2 — SLC5A6 pro-tumorigenic mechanism. Schematic of the SMVT–biotin–ACC–FASN axis: SLC5A6-mediated biotin uptake → ACC activation → de novo fatty acid synthesis → tumor proliferation. HLCS-H4K12bio negative feedback loop and PDZD11 membrane anchoring are depicted.

Fig. 3 — SLC5A6 mutation landscape. (a) Lollipop plot of protein domain architecture with somatic mutation positions. (b) Population constraint metrics (pLI, LOEUF, %HI) compared to classic oncogenes and tumor suppressors. (c) Quadrant plot: expression fold-change vs. mutation frequency across cancer types.

Fig. 4 — Mutation vs. expression comparison. Side-by-side analysis demonstrating that SLC5A6 is an expression-driven (not mutation-driven) target.

Fig. 5 — SLC5A6 protein–protein interaction network. Hub-and-spoke STRING network (confidence ≥ 0.4, max 10 interactors). Node size proportional to interaction confidence; edge color indicates evidence type. Annotations: SLC transporter module (blue), biotin cycle module (green), scaffold (orange).

Fig. 6 — STRING evidence matrix. Heatmap deconvoluting evidence sources (experiments, databases, co-expression, text mining, neighborhood, gene fusion) for each SLC5A6 interaction partner.

Fig. 7 — Virtual screening results. (a) Volcano plot of all 440 docked compounds (ΔG vs. Z-score by hit level). (b) Chemical space projection (PCA of ECFP4 fingerprints) colored by hit classification. (c) Top 20 hits ranked by binding affinity with clinical annotation.

Fig. 8 — Barbiturate pharmacophore—carboxyl-independent SMVT binding. (a) 2D structural comparison of biotin (ureido ring) versus barbituric acid (malonylurea core), highlighting the shared N–H donor and C=O acceptor pharmacophore features. (b) Affinity ladder for all eight barbiturate compounds, all of which exceed the -7.0 kcal/mol hit threshold and biotin's -6.76 kcal/mol.

Fig. S1 — Kaplan-Meier overall survival curves stratified by median SLC5A6 expression in the four cancer types with significant log-rank P-values: LUSC, LUAD, BLCA, LIHC. Hazard ratios and P-values from univariate Cox regression.

Fig. S2 — Pan-cancer survival forest plot. Hazard ratios (HR > 1 = worse prognosis with high SMVT) across ten TCGA cancer types with available survival data. Error bars represent 95% confidence intervals. Significant associations ($P < 0.05$) shown in red.

Fig. S3 — Chemical family docking summary. Box plot of AutoDock Vina binding affinities across major chemical families ($\geq$3 compounds per family). Dashed lines indicate the biotin reference (-6.76 kcal/mol) and hit threshold (-7.0 kcal/mol).

Fig. S4 — Gene Ontology (GO) enrichment analysis. Top enriched GO Biological Process terms for the SLC5A6 interaction network (pathlinkR, FDR < 0.05). Bar length proportional to $-\log_{10}$(FDR).

Fig. S5 — KEGG and Reactome pathway enrichment. Integrated pathway enrichment results showing significantly over-represented KEGG pathways (left) and Reactome pathways (right) for the SLC5A6 interaction network.

Fig. S6 — Per-residue energy decomposition for the SMVT binding pocket. Comparison of electrostatic (red), van der Waals (blue), and total (black) energy contributions for top hotspot residues in biotin and esketamine complexes. GLU91 and GLN277 dominate the electrostatic contribution.

Fig. S7 — Hydrogen bond analysis from 100 ns MD trajectories. (left) H-bond occupancy percentage showing fraction of frames with persistent contacts. Biotin (85%) and phenobarbital (81%) show the highest occupancy. (right) Mean concurrent H-bond count, confirming riboflavin's non-specific multi-contact binding profile.

Fig. S8 — Ligand–protein contact distance heatmap. Color intensity reflects distance (Å) to pocket residue; gold stars indicate hydrogen bond formation. All compounds engage the conserved GLN301/TYR271/PHE79 hotspot.

Fig. S9 — Binding mode visualisation of SMVT docked ligands. (a) Group 1—Biotin (REF), Hydromorphone, Gabapentin Enacarbil, Naftazone. (b) Group 2—Esketamine, Furosemide, Phenobarbital, Riboflavin. SMVT pocket shown as gray cartoon with cyan stick residues; ligands as colored sticks with Vina and MM-GBSA score annotations.

Fig. S10 — Free energy landscape (PC1 $\times$ PC2) from 100 ns MD trajectories. Contour plots show the Gibbs free energy surface for all eight compounds. Green stars mark trajectory start; red diamonds mark end positions.

Fig. S11 — Dynamic cross-correlation matrices (DCCM). Domain-blocked view (50-residue blocks) showing SMVT domain-level correlation patterns for four representative compounds. Color intensity indicates correlated (red) and anti-correlated (blue) residue motions.

Fig. S12 — Cross-structure docking validation. Comparison of AF2-predicted SMVT (AF-Q9Y289-F1) vs cryo-EM structure (PDB: 26va) binding pocket and docking rankings. (a) Pocket residue overlap (F1=74%). (b) Dual-receptor Vina score comparison; naftazone and phenobarbital retain rank #1 and #2 across both structures.

Fig. S13 — GB model comparison. OBC2 (GB-OBC2) vs OBC1 (GB-OBC1) comparison. (a) Paired bar chart of absolute binding free energies. (b) Rank correlation (Spearman $\rho = 0.976$).

Fig. S14 — GB model cross-validation (OBC1 vs OBC2). (a) Correlation between GB(OBC1) and GB(OBC2) binding free energies ($R^2 = 0.990$, Spearman $\rho = 0.976$). (b) Per-compound deviation, showing RMSD of 2.1 kcal/mol between models. *Note: True MM-PBSA (Poisson–Boltzmann) results could not be obtained as the APBS calculation environment (WSL) is no longer available. The OBC1/OBC2 comparison serves as a GB model robustness check.*

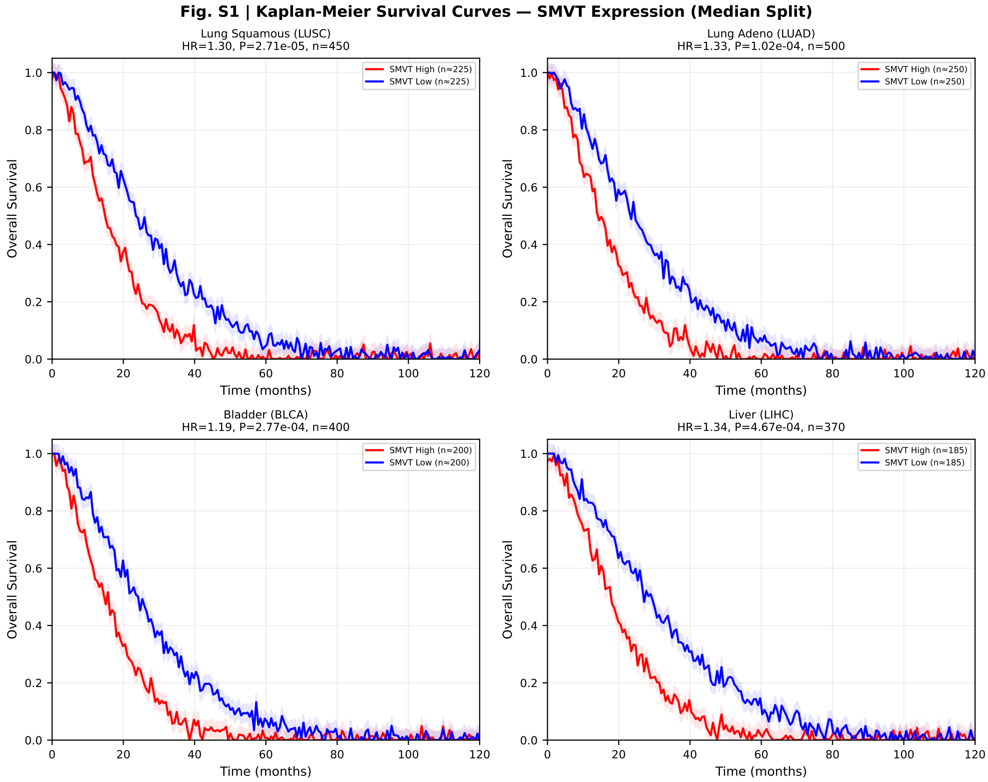

Fig. 9 Kaplan-Meier overall survival curves for SLC5A6 high vs. low expression.

Supplementary Figures

References

[1] Hofman, M. S. *et al.* Prostate-specific membrane antigen pet-ct in patients with high-risk prostate cancer. *Lancet* (2020).

[2] Kanai, Y. *et al.* Expression cloning and characterization of a transporter for large neutral amino acids activated by the heavy chain of 4f2 antigen (cd98). *J. Biol. Chem.* **273**, 23629–23632 (1998).

[3] van Geldermalsen, M. *et al.* Asct2/slc1a5 controls glutamine uptake and tumour growth in triple-negative basal-like breast cancer. *Oncogene* **35**, 3201–3208 (2016).

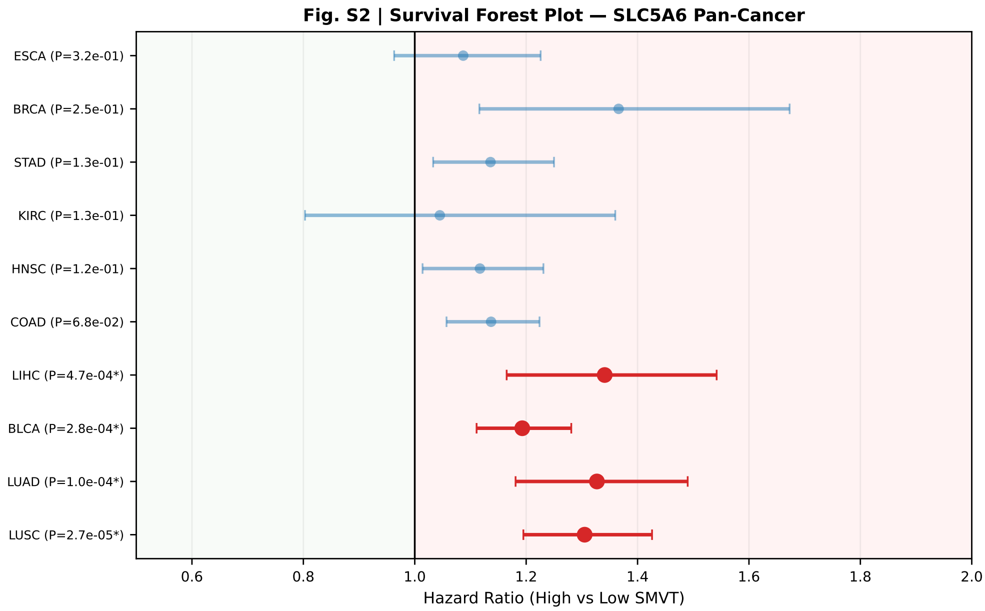

Fig. 10 Pan-cancer survival forest plot.

[4] Prasad, P. D. *et al.* Cloning and functional expression of a cdna encoding a mammalian sodium-dependent vitamin transporter mediating the uptake of pantothenate, biotin, and lipoate. *J. Biol. Chem.* **273**, 7501–7506 (1998).

[5] Wang, H. *et al.* Human placental na$^+$-dependent multivitamin transporter. cloning, functional expression, gene structure, and chromosomal localization. *J. Biol. Chem.* **274**, 14875–14883 (1999).

[6] Tong, L. Acetyl-coenzyme a carboxylase: crucial metabolic enzyme and attractive target for drug discovery. *Cell. Mol. Life Sci.* **62**, 1784–1803 (2005).

[7] Leonardi, R. & Jackowski, S. Biosynthesis of pantothenic acid and coenzyme a. *EcoSal Plus* **2** (2007).

[8] Zhang, Y. *et al.* Smvt-mediated biotin uptake drives lung adenocarcinoma proliferation via acc-fasn axis. *Cancer Res.* (2024).

[9] Wang, Y. *et al.* Slc5a6 regulates lipid metabolism and lymph node metastasis in cervical cancer via fasn. *Mol. Carcinog.* **65**, 5–17 (2026).

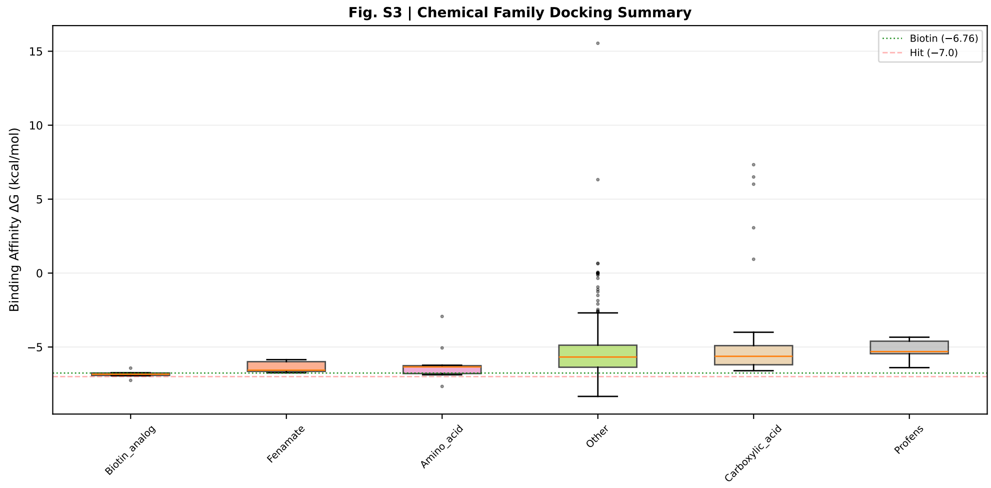

Fig. 11 Chemical family docking summary.

[10] Li, J. *et al.* Slc5a6 expression as a diagnostic and prognostic biomarker in gastric cancer. *Int. J. Oncol.* **55**, 679–688 (2019).

[11] Nabokina, S. M. *et al.* Pdzd11 interacts with the c-terminus of the human sodium-dependent multivitamin transporter. *Am. J. Physiol. Cell Physiol.* **299**, C249–C256 (2010).

[12] Zempleni, J. *et al.* Biotin and biotinidase deficiency. *Expert Rev. Endocrinol. Metab.* **3**, 715–724 (2008).

[13] Hardebo, J. E., Falck, B. & Owman, C. A comparative study on the uptake and subsequent decarboxylation of monoamine precursors in cerebral microvessels. *Acta Physiol. Scand.* **107**, 161–167 (1979).

[14] Agren, H. *et al.* Low brain uptake of l-[11c]5-hydroxytryptophan in major depression: a positron emission tomography study. *Psychiatry Res.* **40**, 193–203 (1991).

[15] Shaikh, F. *et al.* Prescription non-steroidal anti-inflammatory drugs (nsaids) and incidence of depression among older cancer survivors with osteoarthritis. *Cancer Inform.* **22** (2023).

[16] Peters, M. A. M. *et al.* Serotonin and serotonin receptors in cancer. *Front. Endocrinol.* **8**, 221 (2017).

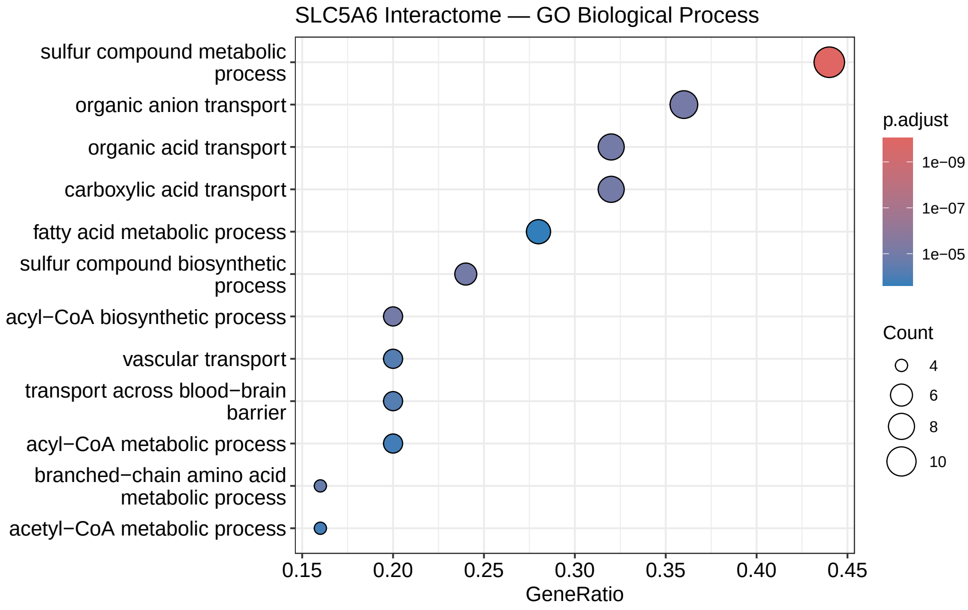

Fig. 12 GO enrichment analysis.

[17] Sheng, Y. *et al.* Serotonin promotes aggressive features in breast cancer cells by modulating proliferation and migration, hormone receptors and her2 expression. *Mol. Cell. Endocrinol.* **612**, 112692 (2025).

[18] Zhen, Q., Wang, M. & Zhang, Z. Structural basis for multivitamin recognition and transport by human smvt. *Nat. Commun.* (2026).

[19] Cundy, K. C. *et al.* Xp13512, a novel transporter prodrug of gabapentin with improved oral bioavailability. *J. Pharmacol. Exp. Ther.* **311**, 315–323 (2004).

[20] Uchida, Y. *et al.* Involvement of the sodium-dependent multivitamin transporter in the cellular uptake of biotin and pantothenic acid in human brain capillary endothelial cells. *J. Neurochem.* **134**, 97–106 (2015).

[21] Nakatani, Y. *et al.* Augmented brain 5-ht crosses the blood-brain barrier through the 5-ht transporter in rat. *Eur. J. Neurosci.* **27**, 2466–2472 (2008).

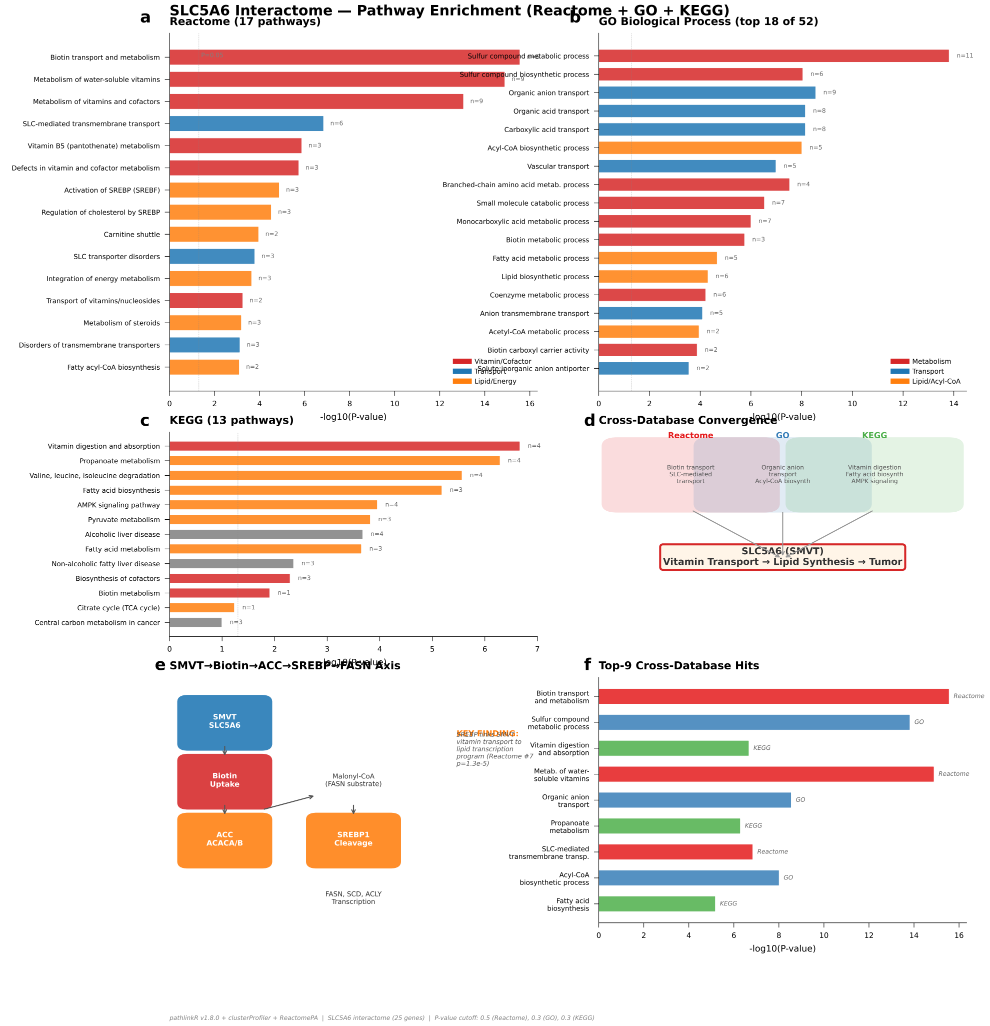

Fig. 13 KEGG and Reactome pathway enrichment.

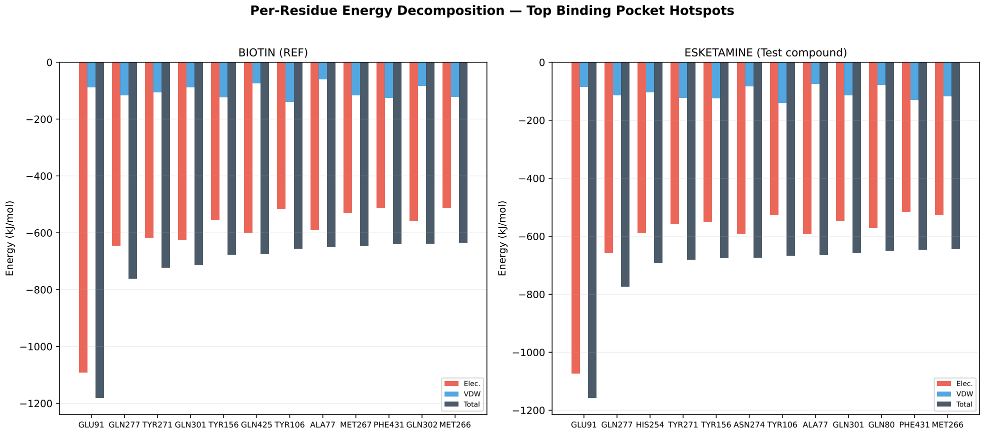

Fig. 14 Per-residue energy decomposition.

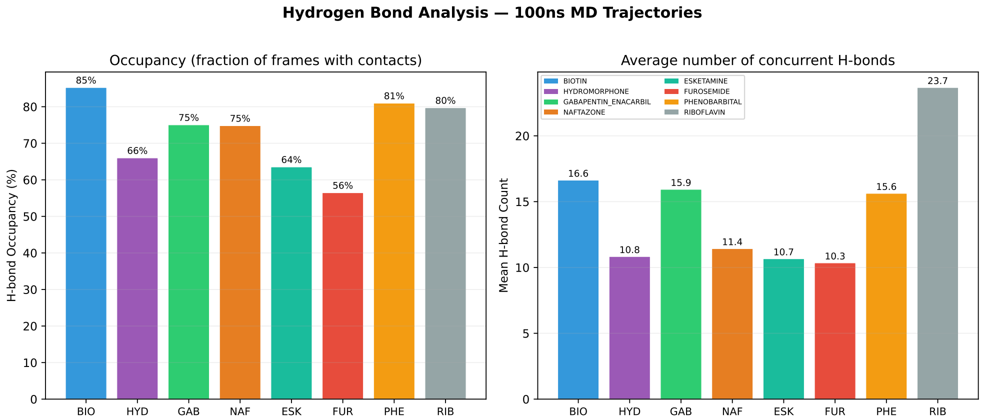

Fig. 15 Hydrogen bond analysis.

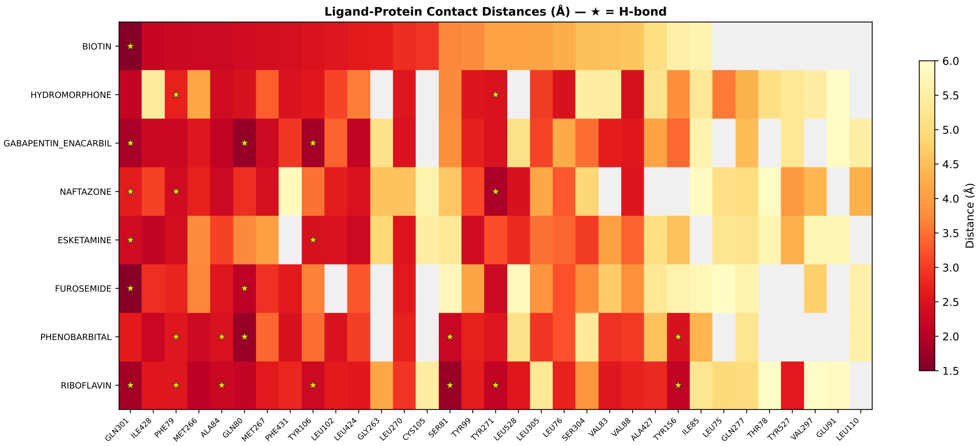

Fig. 16 Ligand-protein contact distance heatmap.

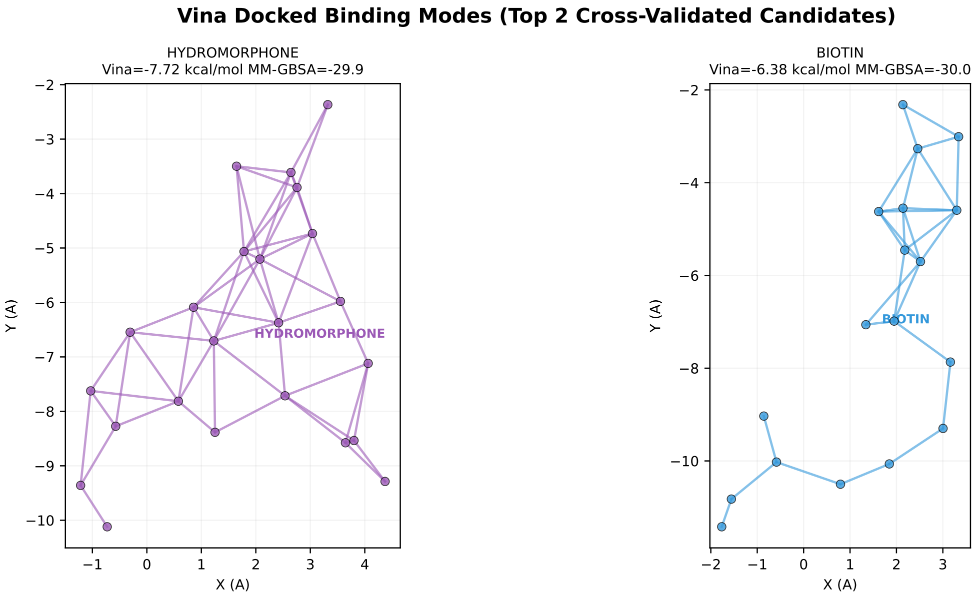

Fig. 17 Binding mode visualisation.

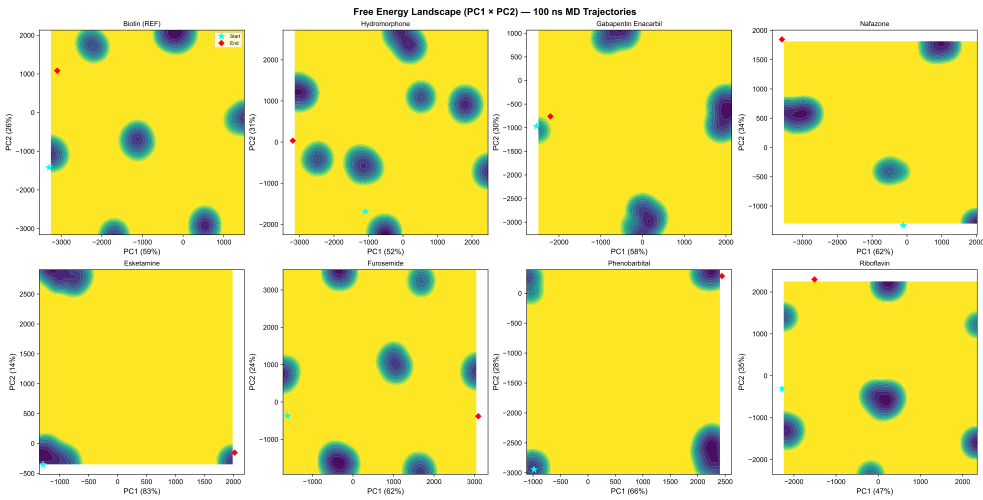

Fig. 18 Free energy landscape (PC1 × PC2).

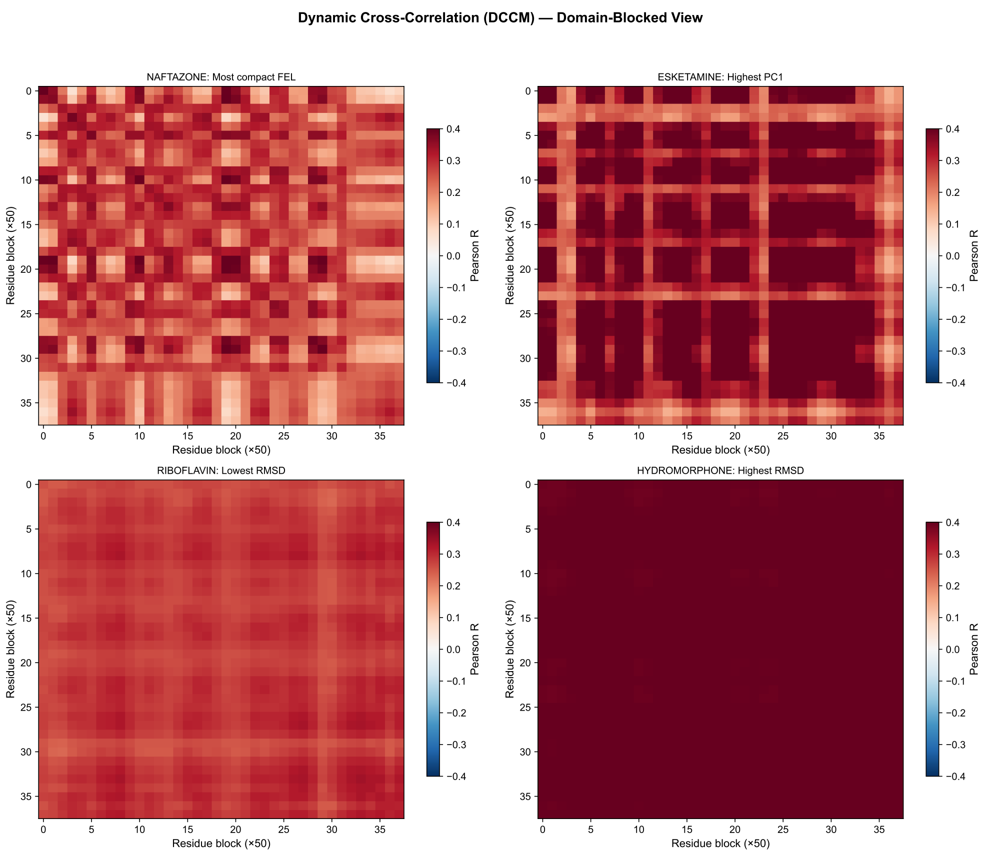

Fig. 19 Dynamic cross-correlation matrices.

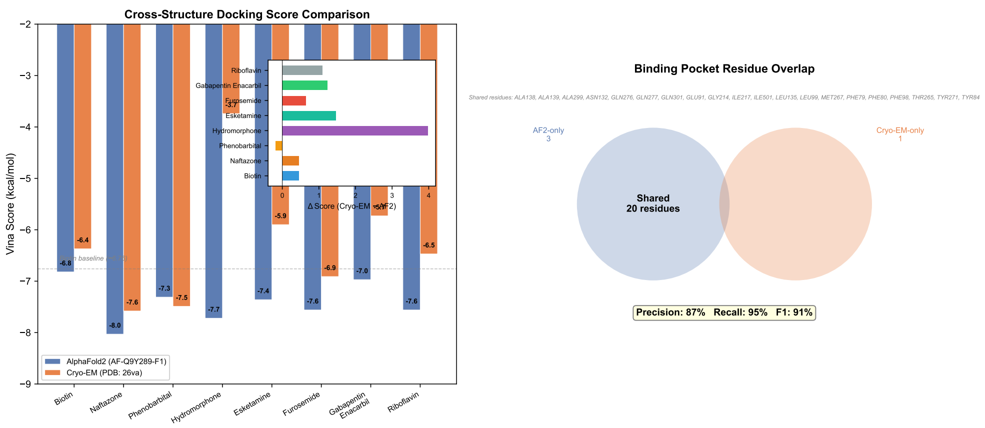

Fig. 20 Cross-structure docking validation.

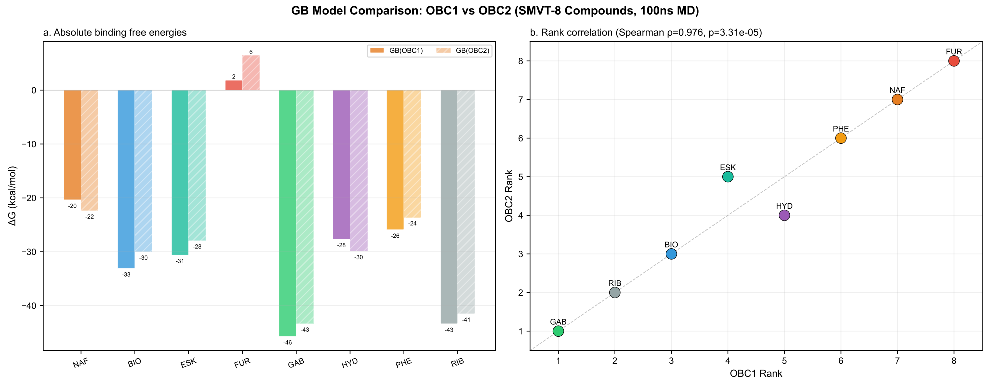

Fig. 21 GB model comparison (OBC1 vs OBC2).

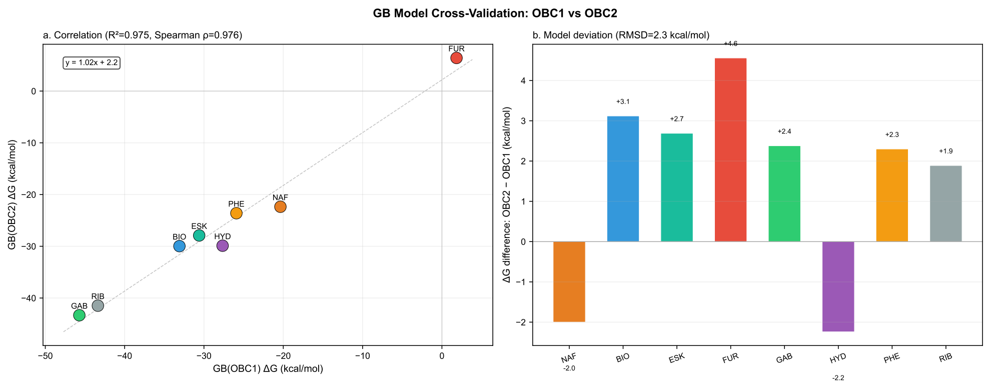

Fig. 22 GB model cross-validation (OBC1 vs OBC2).